Fiscal Year:	FY 2008	**Task Last Updated:**	FY 06/18/2008
PI Name:	Chapes, Stephen K. Ph.D.		
Project Title:	Differentiation of Bone Marrow Macrophages in Space		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	None		
PI Email:	skcbiol@ksu.edu	**Fax:**	FY 785-532-6653
PI Organization Type:	UNIVERSITY	**Phone:**	785-532-6795
Organization Name:	Kansas State University		
PI Address 1:	1717 Claflin Rd.		
PI Address 2:	Division of Biology		
PI Web Page:			
City:	Manhattan	**State:**	KS
Zip Code:	66506-4901	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	96-HEDS-04
Start Date:	12/28/1998	**End Date:**	03/31/2008
No. of Post Docs:	2	**No. of PhD Degrees:**	1
No. of PhD Candidates:	3	**No. of Master' Degrees:**	2
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	Shuttle/ISS		
Flight Assignment:	STS-118 ; STS-115 NOTE: Received NCE to 3/31/2008 per A. Chu/ARC (7/07) NOTE: Received NCE through 9/30/2007, per A. Chu (9/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG2-1274		
Performance Goal No.:			
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Task Description:	Space flight and/or microgravity have dramatic effects on macrophage and other cells. For example, murine bone marrow macrophages respond to microgravity in a period of about 8 seconds. These early responses to microgravity are associated with observations that bone marrow stem cell differentiation into mature macrophages is inhibited during space flight. The specific aim of this project is to determine mechanisms for the disrupted macrophage differentiation during space flight. I propose to test two hypotheses to address this specific aim: (1) Macrophage differentiation is affected because the receptor for CSF-1, c-fms is down regulated on macrophage precursors, and (2) Space flight disrupts c-fos-dependent processes involved in macrophage differentiation.We will use SHOT Inc. hardware, the Cellcult unit and bioreactor to grow primary bone marrow in the presence of macrophage colony stimulating factor. All experimental preparations will be done at the PI's home institution or at Morehouse University in Atlanta. Therefore, this project meets the requirement for needing limited crew training and involvement as described in NRA 96-HEDS-04. Moreover, this project has a flight history and a sound experimental rationale to justify its execution. Due to the changes in NASA priorities, this project was terminated and in the final grant period, our group has developed and tested tools that can be used for future shuttle missions. Indeed, some of the assays we developed over the last year will be used to assess bone marrow samples that will be obtained from a cooperative agreement with a commercial flight experiment to be done on STS-118 (Commercial Biomedical Test Module-2, CBTM-2).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Space flight offers a unique environment that impacts many physiological systems. Amongst these systems, bone metabolism is known to be significantly altered by the skeletal unloading of space flight. Hematopoiesis is the process which provides the body with mature, functioning cells necessary for oxygenation, clotting and immune functions. The bone, specifically, the bone marrow, is the site for hematopoiesis.Moreover, changes in bone impact all compartments therein, including the marrow. Indeed, analysis of animals from rats to rhesus moneky to humans shows that hematopoiesis can be affected by space flights. Although direct examination of primary hematopoiesis is not an aim of the proposed project, this experiment will allow us to dissect a very complex system. We will obtain valuable information about what mechanisms involved in hematopoiesis are susceptible to disruption by skeletal unloading. This understanding will allow for more effective countermeasure planning. On Earth, skeletal unloading occurs to individuals that are bedridden for long periods of time, wheel-chair bound, or lose the ability to stand for long periods of time. These prolonged periods of skeletal unloading also induce changes in bone that parallel those seen in space flight. In addition, individuals with bone disorders, such as osteoporosis, are also subject to alterations in hematopoiesis. This is best exemplified by the osteopetrotic mouse, which has diminished numbers of macrophages and brittle bones. Therefore, data obtained from this study will also have direct extrapolations to individuals on Earth.
Task Progress:	Bone marrow molecular surface expression was examined as part of the Commercial Biomedical Test Module-2 (CBTM-2) payload experiment. Bone marrow cells were isolated from the humerii of placebo-treated, C57BL/6J mice following an 11-day flight on STS-118. Bone marrow cells were analyzed as complete bone marrow and cell subpopulations were identified based on size and granularity and analyzed. Ly6C was the only surface molecule significantly elevated on whole bone marrow cells from Flight mice compared to AEM-housed mice. This contrasts the observations of Pecaut et al. (Pecaut, 2003) who did not see changes in Ly6A/E in their analysis of mouse marrow following a 12 day flight on STS-108. This difference is probably because Ly6A/E is expressed on different cell populations than Ly6C (Barclay, 1993). This hypothesis is supported by the observations from STS-108 (Pecaut, 2003) and STS-118 where there were no differences in numbers of bone marrow cells isolated from Flight and AEM-housed mice. Although Ly6C expression was elevated on whole bone marrow isolated from flight mice compared to AEM controls, when the data set was assessed as a whole (all phenotypes considered side-by-side), there was not a significant difference between flight and bone marrow mice (P>0.2 by chi squared). This was not surprising because bone marrow contains a heterogeneous population of cells that share phenotypes. Therefore, we analyzed subpopulations of cells that were distinguished on their size and granularity. Three subpopulations of cells were identified and were assessed for the expression of Ly6C, CD11b, CD31, Ly6G, F4/80, CD44 and cfos. There were subpopulation differences in the change in expression between space flight. For example, Ly6C expression was elevated by space flight in the small agranular R1 subpopulation but was lower in the R3 subpopulation. The R1 population contained the fewest numbers of macrophages and neutrophils based on the expression of F4/80 and Ly6G. Given that Ly6C is expressed on cells at a less mature stage within the myeloid lineage (Leenen, 1994), these data suggest that the impact of space flight was being observed on less mature cells compared to more mature cells in the granulocytic lineage. CD11b was significantly elevated on bone marrow cells in the R2 population isolated from flight mice compared to AEM-housed mice. CD11b is involved in trafficking and strong adhesion of macrophages and granulocytes in trafficking. Therefore, the elevation of this marker on the R2 subpopulation but not R1 or R3 subpopulation may reflect the activation of the enriched neutrophil population in the R2 group in response to landing. Neutrophilia has been seen as a consequence of landing stress in past rodent flights (Allebban, 1994, Chapes, 1999, Chapes, 1999, Ichiki, 1996). Therefore, the specific impact on the R2 subpopulation would be consistent with those observations. The major decrease in high level expression of CD44 and Ly6G in the R3 bone marrow subpopulation from Flight mice but not R1 or R3 subpopulations was also notable. In fact, there was a reverse correlation between the high-level expression of CD44 and Ly6G, which are strongly expressed on PMNs and the increased expression of F4/80 in R3. Because F4/80 identifies macrophages these data suggest that flight is increasing the percentage of macrophages in the bone marrow or decreasing the percentages of PMNs. The stress of landing could induce the mobilization of bone marrow PMNs into circulation and cause such and impact (Allebban, 1994, Chapes, 1999; Chapes, 1999; Ichiki,). Because the shift in cellular phenotypes is a complex process. We analyzed the complete data set using both parametric (Factorial ANOVA) and nonparametric statistics (chi squared). R1-R3 comparisons between Flight and AEM-house mice suggests that as a whole there are significant differences (P<0.01) with either type of analysis. The mice were analyzed post-landing. Therefore, it is not possible to separate the landing effects from the flight effects. However, the data overall suggests that there are significant changes in bone marrow phenotype in response to the space flight

	experience. It is not clear if these changes translate into changes in immune competence {Chapes, 2005 #10500}, but the data are consistent with other observations that bone marrow changes occur during space flight (Sonnenfeld, 1990; Vacek, 1983; Sonnenfeld, 1992; Ichiki, 1996).
Bibliography Type:	Description: (Last Updated: 09/15/2021)
Articles in Peer-reviewed Journals	Potts BE, Chapes SK. "Functions of C2D macrophage cells after adoptive transfer." J Leukoc Biol. 2008 Mar;83(3):602-9. Epub 2007 Dec 6. PMID: 18063699 , Mar-2008
Articles in Peer-reviewed Journals	Potts BE, Hart ML, Snyder LL, Boyle D, Mosier DA, Chapes SK. "Differentiation of C2D macrophage cells after adoptive transfer." Clin Vaccine Immunol. 2008 Feb;15(2):243-52. Epub 2007 Dec 19. PMID: 18094115 , Feb-2008
Articles in Peer-reviewed Journals	Xie L, O'Reilly CP, Chapes SK, Mora S. "Adiponectin and leptin are secreted through distinct trafficking pathways in adipocytes." Biochim Biophys Acta. 2008 Feb;1782(2):99-108. Epub 2008 Jan 7. PMID: 18179777 , Feb-2008
Articles in Peer-reviewed Journals	Luce-Fedrow A, Von Ohlen T, Boyle D, Ganta RR, Chapes SK. "Use of Drosophila S2 cells as a model for studying Ehrlichia chaffeensis infections." Appl Environ Microbiol. 2008 Mar;74(6):1886-91. Epub 2008 Feb 1. PMID: 18245255 , Mar-2008
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